

88AMO4

SomeonecoolLovesMaths

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Problem

Let I be the incenter of triangle ABC , and let A' , B' , and C' be the circumcenters of triangles IBC , ICA , and IAB , respectively. Prove that the circumcircles of triangles ABC and $A'B'C'$ are concentric.

Solution. Due to the incenter-excenter lemma, A' is the intersection of AI with (ABC) . Similarly, B', C' lie on (ABC) finishing the problem.